Problem 1/20

JOIN / IS NULL

AMAZON

No Sales

Find employees who have never made a sale.

SAMPLE INPUT

EMP_ID	NAME	SALE_ID
1	Alice	101
2	Bob	NULL

EXPECTED OUTPUT

ID	NAME
2	Bob

SOLUTION

MYSQL / SQL SERVER

```sql
-- Identify rows with no matching join
SELECT e.id, e.name
FROM employees e
LEFT JOIN sales s ON e.id = s.emp_id
WHERE s.sale_id IS NULL;
```

Next: Date Math →

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Problem 2/20

DATE MATH

GOOGLE

Avg Tenure

Calculate average years of service per department.

SAMPLE INPUT

DEPT	HIRED
Eng	2020-01-01
Eng	2022-01-01

EXPECTED OUTPUT

DEPT_ID	AVG_YEARS
Eng	3

SOLUTION

MYSQL / SQL SERVER

```sql
-- Calculate diff in years from today
SELECT dept_id,
  AVG(TIMESTAMPDIFF(YEAR, hire_date, CURDATE()))
  AS avg_tenure
FROM employees
GROUP BY dept_id;
```

Next: NTILE Window →

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Problem 3/20 NTILE

MICROSOFT

Top Earners

Get employees in the top 10% salary bracket of their dept.

SAMPLE INPUT

NAME	SALARY
Alice	200k
Bob	50k

EXPECTED OUTPUT

NAME	DECILE
Alice	1

SOLUTION

SQL SERVER

```sql
-- Divide dept into 10 buckets
SELECT * FROM (
  SELECT *, NTILE(10) OVER(
    PARTITION BY dept_id ORDER BY salary DESC
  ) as decile
  FROM employees
) sub
WHERE decile = 1;
```

Next: Fraud Detection →

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SAVE REPOST

Problem 4/20

GROUP BY

META

Fraud Check

Find customers who purchased more than once in the same day.

SAMPLE INPUT

CUST_ID	DATE
C1	2024-01-01
C1	2024-01-01

EXPECTED OUTPUT

CUST_ID	DATE
C1	2024-01-01

SOLUTION

MYSQL

```sql
-- Filter groups with count > 1
SELECT customer_id, purchase_date
FROM sales
GROUP BY customer_id, purchase_date
HAVING COUNT(*) > 1;
```

Next: Reporting →

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Problem 5/20

LEFT JOIN

NETFLIX

Dept Counts

List all departments and their employee counts (including zero).

SAMPLE INPUT

DEPT	EMP_ID
HR	101
Legal	NULL

EXPECTED OUTPUT

DEPT	COUNT
HR	1
Legal	0

SOLUTION

MYSQL / SQL SERVER

```sql
-- Count on employee id handles NULLs
SELECT d.dept_name, COUNT(e.id)
FROM departments d
LEFT JOIN employees e ON d.id = e.dept_id
GROUP BY d.dept_name;
```

Next: Data Quality →

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Problem 6/20

AGGREGATE

APPLE

Duplicates

Find duplicate rows based on multiple columns.

SAMPLE INPUT

COL1	COL2
A	X
A	X
B	Y

EXPECTED OUTPUT

COL1	COL2	CNT
A	X	2

SOLUTION

MYSQL / SQL SERVER

```sql
-- Group by all columns to find dupes
SELECT col1, col2, COUNT(*)
FROM table_name
GROUP BY col1, col2
HAVING COUNT(*) > 1;
```

Next: Recursive CTE →

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REPOST

Problem 7/20

RECURSIVE

UBER

Factorial

Calculate factorial (5!) using a Recursive CTE.

SAMPLE INPUT

N (LIMIT)
5

EXPECTED OUTPUT

FACT
120

SOLUTION

SQL SERVER / MYSQL 8+

```sql
-- Recursive multiplication
WITH RECURSIVE Factorial(n, fact) AS (
  SELECT 1, 1
  UNION ALL
  SELECT n+1, fact*(n+1) FROM Factorial WHERE n < 5
)
SELECT fact FROM Factorial WHERE n=5;
```

Next: Cumulative Sales →

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SAVE

REPOST

Problem 8/20

WINDOW FUNC

AIRBNB

Running %

Calculate cumulative percentage of total sales per product.

SAMPLE INPUT

ID	AMT
P1	100
P2	300

EXPECTED OUTPUT

ID	CUM_PCT
P2	75.0
P1	100.0

SOLUTION

SQL SERVER / MYSQL 8+

```sql
-- Sum Over Order / Sum Over All
SELECT id, amt,
  SUM(amt) OVER(ORDER BY amt DESC) * 100.0 /
  SUM(amt) OVER() as cum_pct
FROM sales;
```

Next: Org Hierarchy →

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REPOST

Problem 9/20

RECURSIVE

INFOSYS

Manager Chain

Find all employees reporting directly or indirectly to a manager.

SAMPLE INPUT

ID	MGR_ID
101	NULL
102	101
103	102

EXPECTED OUTPUT

ID	MGR
102	101
103	102

SOLUTION

SQL SERVER / MYSQL 8+

```sql
-- Traverse hierarchy downwards
WITH RECURSIVE Hierarchy AS (
  SELECT id, manager_id FROM employees WHERE manager_id = 101
  UNION ALL
  SELECT e.id, e.manager_id
  FROM employees e JOIN Hierarchy h ON e.manager_id = h.id
)
SELECT * FROM Hierarchy;
```

Next: Statistics →

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SAVE

REPOST

Problem 10/20 STATS

TCS

Order Stats

Calculate avg orders per customer with standard deviation.

SAMPLE INPUT

CUST	ORDERS
A	2
B	4

EXPECTED OUTPUT

AVG	STDDEV
3.0	1.41

SOLUTION

MYSQL / SQL SERVER

```sql
-- CTE first for counts
WITH OrderCounts AS (
  SELECT cust_id, COUNT(*) as cnt
  FROM orders GROUP BY cust_id
)
SELECT AVG(cnt), STDDEV(cnt) FROM OrderCounts;
```

Next: Missing Data →

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SAVE REPOST

Problem 11/20

LEAD()

ACCENTURE

Missing Dates

Find gaps in date sequences (Missing Days).

SAMPLE INPUT

DATE
2024-01-01
2024-01-03

EXPECTED OUTPUT

DATE	MISSING
2024-01-01	2024-01-02

SOLUTION

SQL SERVER / MYSQL 8+

```sql
-- Compare current date + 1 with next date
SELECT DATE_ADD(date, INTERVAL 1 DAY) as missing
FROM (
  SELECT date, LEAD(date) OVER(ORDER BY date) as next_d
  FROM sales
) sub
WHERE next_d > DATE_ADD(date, INTERVAL 1 DAY);
```

Next: Ranking Logic →

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SAVE

REPOST

Problem 12/20

RANKING

ADOBE

Rank Types

Rank employees by salary using RANK vs PERCENT_RANK.

SAMPLE INPUT

SALARY
100k
80k

EXPECTED OUTPUT

RNK	PCT_RNK
1	0
2	1

SOLUTION

MYSQL / SQL SERVER

```sql
-- Compare ranking methods
SELECT name, salary,
  RANK() OVER(ORDER BY salary DESC) as rnk,
  PERCENT_RANK() OVER(ORDER BY salary DESC) as pct
FROM employees;
```

Next: Not Exists →

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SAVE

REPOST

Problem 13/20

NOT EXISTS

SALESFORCE

Unsold Items

Find products that have NEVER been sold.

SAMPLE INPUT

PROD_ID	SOLD
A	Yes
B	No

EXPECTED OUTPUT

PROD_NAME
Product B

SOLUTION

MYSQL / SQL SERVER

```sql
-- Correlated subquery for existence
SELECT p.product_name
FROM products p
WHERE NOT EXISTS (
  SELECT 1 FROM sales s WHERE s.product_id = p.id
);
```

Next: Gaps & Islands →

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SAVE

REPOST

Problem 14/20

GAPS/ISLANDS

LINKEDIN

Sales Streaks

Find consecutive days where sales were above $1000.

SAMPLE INPUT

DATE	AMT
Jan 1	1200
Jan 2	1500
Jan 3	500

EXPECTED OUTPUT

START	END	DAYS
Jan 1	Jan 2	2

SOLUTION

SQL SERVER / MYSQL 8+

```sql
-- Group by Date minus Row Number
SELECT MIN(date), MAX(date), COUNT(*)
FROM (
  SELECT date, ROW_NUMBER() OVER(ORDER BY date) as rn
  FROM sales WHERE amount > 1000
) t
GROUP BY DATE_ADD(date, INTERVAL -rn DAY);
```

Next: String Agg →

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SAVE

REPOST

Problem 15/20

GROUP_CONCAT

FLIPKART

Concatenate

Create comma-separated list of employee names per dept.

SAMPLE INPUT

DEPT	NAME
IT	Alice
IT	Bob

EXPECTED OUTPUT

DEPT	EMPLOYEES
IT	Alice, Bob

SOLUTION

MYSQL

```sql
-- Aggregation into string
SELECT department_id,
  GROUP_CONCAT(name SEPARATOR ', ') as employees
FROM employees
GROUP BY department_id;
```

Next: Window Filter →

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SAVE

REPOST

Problem 16/20

WINDOW AVG

SWIGGY

Salary Band

Find employees earning > Dept Avg but < Comp Avg.

SAMPLE INPUT

SAL	DEPT_AVG	COMP_AVG
60k	50k	70k

EXPECTED OUTPUT

NAME
Candidate A

SOLUTION

MYSQL / SQL SERVER

```sql
-- Filter based on two window averages
SELECT * FROM (
  SELECT e.*,
    AVG(salary) OVER(PARTITION BY dept) as dept_avg,
    AVG(salary) OVER() as comp_avg
  FROM employees e
) t
WHERE salary > dept_avg AND salary < comp_avg;
```

Next: Set Theory →

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SAVE

REPOST

Problem 17/20 | HAVING | ZOMATO

Complete Set

Customers who purchased ALL products in category 10.

SAMPLE INPUT

CUST	PRODS_BOUGHT
A	3
Total_Prods	3

EXPECTED OUTPUT

CUSTOMER_ID
A

SOLUTION

MYSQL / SQL SERVER

```sql
-- Compare count distinct with total count
SELECT customer_id
FROM sales WHERE cat_id = 10
GROUP BY customer_id
HAVING COUNT(DISTINCT prod_id) = (
  SELECT COUNT(*) FROM products WHERE cat_id = 10
);
```

Next: Nth Highest →

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SAVE REPOST

Problem 18/20

OFFSET

STRIPE

Nth Highest

Retrieve the Nth highest salary (Generic solution).

SAMPLE INPUT

SALARY
100
90
80

EXPECTED OUTPUT

3RD_HIGHEST
80

SOLUTION

MYSQL / SQL SERVER

```sql
-- Limit 1 Offset N-1
SELECT DISTINCT salary
FROM employees
ORDER BY salary DESC
LIMIT 1 OFFSET 2; -- For 3rd highest
```

Next: Audit Join →

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The 20-Pattern SQL Cheat Sheet

SAVE

REPOST

Problem 19/20

ANTI JOIN

ATLASSIAN

Missing Logs

Find employees with no salary history entries.

SAMPLE INPUT

EMP	HISTORY_ID
Alice	101
Bob	NULL

EXPECTED OUTPUT

NAME
Bob

SOLUTION

MYSQL / SQL SERVER

```sql
-- Left Join where right is null
SELECT e.*
FROM employees e
LEFT JOIN salary_history sh ON e.id = sh.emp_id
WHERE sh.emp_id IS NULL;
```

Next: Basic Stats →

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SAVE

REPOST

Problem 20/20

ORDER BY

SNOWFLAKE

Largest Dept

Find which department has the most employees.

SAMPLE INPUT

DEPT	COUNT
Sales	50
IT	30

EXPECTED OUTPUT

DEPT	CNT
Sales	50

SOLUTION

MYSQL / SQL SERVER

```sql
-- Order by count desc limit 1
SELECT department_id, COUNT(*) as cnt
FROM employees
GROUP BY department_id
ORDER BY cnt DESC
LIMIT 1;
```

Next: MISSION ACCOMPLISHED →

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REPOST

MISSION ACCOMPLISHED

You're Interview Ready!

20 PATTERNS MASTERED

- ✓ JOIN / IS NULL
- ✓ DATE MATH
- ✓ NTILE
- ✓ GROUP BY
- ✓ LEFT JOIN
- ✓ AGGREGATE
- ✓ RECURSIVE
- ✓ WINDOW FUNC
- ✓ RECURSIVE
- ✓ STATS
- ✓ LEAD()
- ✓ RANKING
- ✓ NOT EXISTS
- ✓ GAPS/ISLANDS
- ✓ GROUP_CONCAT
- ✓ WINDOW AVG
- ✓ HAVING
- ✓ OFFSET
- ✓ ANTI JOIN
- ✓ ORDER BY
- ♥ Follow

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